

The Real Theory of Commerce

Exchange, Persistence, Loss, Insurance, and Optimal Stopping

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

Commerce is conventionally represented as exchange: goods, services, capital, or claims move between economic agents in anticipation of reciprocal value. This representation is incomplete. A commercial system simultaneously realizes exchange, establishes persistent productive relationships, exposes capital to destruction, prices risk through insurance, and generates a sequential decision problem concerning whether commerce should continue.

This treatise develops a compact interdisciplinary theory around a canonical ten-ship representation: eight ships return with added value, one remains abroad to sustain continuing commerce, and one is lost. The identity

$$10 = 8 + 1 + 1 \quad (1)$$

is not proposed as an empirical universal ratio. It is a structural representation of three distinct fates of commercial capital: realization, persistence, and destruction.

The framework connects international trade, capital accumulation, insurance, optimal stopping, stochastic control, welfare economics, econometrics, network theory, evolutionary biology, psychology, political economy, diplomacy, geopolitics, warfare economics, computer science, and artificial intelligence. The central conclusion is that commerce is best understood as a stochastic institution for transforming resources through exchange while simultaneously managing persistence and irreducible risk.

The treatise ends with “The End”

Contents

1	Introduction	2
2	The Ten-Ship Representation	3
3	Commerce as Value Transformation	3
4	The Persistence Principle	4
5	The Loss Principle and Insurance	4
6	Commerce and Optimal Stopping	5
7	The Commercial Continuation Premium	5
8	Risk, Return, and the Commercial Decision	5
9	Econometric Interpretation	6
10	Statistics and Information	6
11	Psychology and Behavioral Economics	6
12	Biology and Evolution	7
13	Physics and Thermodynamics	7

14 Chemistry	7
15 Network Theory	7
16 International Relations and Diplomacy	8
17 Geopolitics	8
18 Warfare Economics	8
19 Welfare Economics	8
20 Political Economy	9
21 Philosophy	9
22 Religion and Moral Economy	9
23 Computer Science and Algorithms	9
24 Machine Learning	10
25 Data Science Architecture	10
26 A Unified Commercial Architecture	11
27 The Core Propositions	11
28 A Compact Accounting Table	12
29 The General Commercial Equation	12
30 Limits of the Theory	12
31 Conclusion	12

List of Figures

1 The commercial system as exchange, persistence, loss, insurance, and sequential decision-making.	11
--	----

List of Tables

1 The canonical ten-ship commercial system	12
--	----

1 Introduction

Commerce predates modern economic theory. Exchange appears in anthropological accounts of reciprocity and redistribution, in classical political economy, and in modern theories of international trade, contracts, institutions, and networks [30, 27, 24, 22, 18].

The usual abstraction is a bilateral exchange:

$$A \longleftrightarrow B. \quad (2)$$

The present theory adds three dimensions. First, commercial capital may return in an enhanced form. Second, some capital may remain in the foreign economy and establish continuing productive capacity. Third, some capital may be destroyed.

Thus a commercial expedition is not simply an exchange transaction. It is a stochastic transformation:

$$\text{Initial Commercial Capital} \longrightarrow \text{Return, Persistence, Loss}. \quad (3)$$

The distinction is important because persistent foreign capital is economically different from destroyed capital. One does not return because it has been transformed into a continuing foreign commercial position; the other does not return because it has ceased to exist as productive capital.

2 The Ten-Ship Representation

Suppose a nation dispatches ten ships, each carrying commercial value v . Initial commercial value is

$$V_0 = 10v. \quad (4)$$

The canonical allocation is

$$10 = 8 + 1 + 1. \quad (5)$$

Let

$$N_R = 8, \quad (6)$$

$$N_F = 1, \quad (7)$$

$$N_L = 1, \quad (8)$$

where R denotes returning ships, F persistent foreign capital, and L loss.

More generally, for a fleet of N ships,

$$N = N_R + N_F + N_L. \quad (9)$$

Define the corresponding fractions:

$$r = \frac{N_R}{N}, \quad (10)$$

$$f = \frac{N_F}{N}, \quad (11)$$

$$\ell = \frac{N_L}{N}. \quad (12)$$

Consequently,

$$r + f + \ell = 1. \quad (13)$$

The ten-ship representation gives

$$r = 0.8, \quad f = 0.1, \quad \ell = 0.1. \quad (14)$$

These numbers constitute a benchmark, not a statistical assertion.

3 Commerce as Value Transformation

Let each returning ship have value $(1 + g)v$, where g is the average gross commercial return.

Then

$$V_R = 8(1 + g)v. \quad (15)$$

The foreign commercial asset has initial value

$$V_F = v, \quad (16)$$

while physical loss is

$$V_L = v. \quad (17)$$

The economically relevant distinction is therefore

$$V_F \neq V_L. \quad (18)$$

Both represent one ship in the physical decomposition, but only V_F remains productive.

This distinction generalizes to modern international investment. A foreign factory, warehouse, vessel, branch, financial claim, or distribution network may cease to be domestic physical capital without becoming economic loss. It becomes part of a cross-border capital structure [13, 9].

4 The Persistence Principle

Let K_t denote foreign commercial capital accumulated through previous voyages. Then

$$K_{t+1} = K_t + V_{F,t} - V_{F,t}^{\text{withdrawn}}. \quad (19)$$

Persistent foreign capital creates a network through which future transactions become possible.

If a foreign asset of value v generates a probability q of supporting another commercial cycle, its simple discounted continuation value can be represented as

$$V_F^* = v + \beta q v + (\beta q)^2 v + \dots. \quad (20)$$

For $0 < \beta q < 1$,

$$V_F^* = \frac{v}{1 - \beta q}. \quad (21)$$

Thus a ship remaining abroad may have a capitalized economic value substantially exceeding its immediate physical value.

This is consistent with the broader economic principle that capital is valued partly by the stream of future services it produces [10, 12].

5 The Loss Principle and Insurance

A commercial system requires a distinction between deterministic loss and probabilistic risk.

Let p denote the probability of ship loss:

$$p = P(L > 0). \quad (22)$$

A non-degenerate insurance market requires

$$0 < p < 1. \quad (23)$$

If

$$p = 0, \quad (24)$$

then the modeled maritime loss is impossible and the associated insurance risk disappears.

Insurance does not eliminate physical loss. It redistributes its financial consequences across a risk pool. Under a simplified actuarial representation, expected loss is

$$E[L] = Npv. \quad (25)$$

An actuarially fair premium can be represented schematically as

$$\Pi = E[L] + \text{administrative cost} + \text{capital charge}, \quad (26)$$

subject to loading for uncertainty and insurer capital requirements [2, 29].

The canonical ten-ship system has

$$p = \frac{1}{10} \quad (27)$$

if interpreted as an average loss rate of one ship per ten ships.

The crucial theoretical point is not that exactly one ship must sink on every voyage. Rather,

$$P(L > 0) > 0 \quad (28)$$

must hold for maritime loss insurance to represent a non-degenerate risk-pooling problem.

6 Commerce and Optimal Stopping

Commerce is sequential. After every voyage, agents decide whether to undertake another voyage.

Let the commercial state be

$$X_t = (K_t, V_t, R_t, L_t), \quad (29)$$

where K_t is persistent foreign capital, V_t realized commercial wealth, R_t the risk state, and L_t accumulated loss.

Let $S(X_t)$ denote the liquidation or stopping value and $C(X_t)$ the value of continuing.

The value function satisfies the Bellman equation

$$W(x) = \max \{S(x), E_x [R(X_{t+1}) + \beta W(X_{t+1})]\}. \quad (30)$$

The optimal stopping time is

$$\tau^* = \inf \{t : S(X_t) \geq E [R(X_{t+1}) + \beta W(X_{t+1}) \mid X_t]\}. \quad (31)$$

Thus the merchant continues while

$$C(X_t) > S(X_t) \quad (32)$$

and stops when

$$C(X_t) \leq S(X_t). \quad (33)$$

This connects the theory directly to optimal stopping and stochastic control [4, 26].

7 The Commercial Continuation Premium

Define

$$\Gamma(X_t) = C(X_t) - S(X_t). \quad (34)$$

Then

$$\Gamma(X_t) > 0 \quad \Rightarrow \quad \text{continue}, \quad (35)$$

$$\Gamma(X_t) = 0 \quad \Rightarrow \quad \text{indifference}, \quad (36)$$

$$\Gamma(X_t) < 0 \quad \Rightarrow \quad \text{stop}. \quad (37)$$

The stopping boundary is therefore

$$\mathcal{S} = x : C(x) \leq S(x), \quad (38)$$

while the continuation region is

$$\mathcal{C} = x : C(x) > S(x). \quad (39)$$

Commerce becomes a free-boundary problem.

8 Risk, Return, and the Commercial Decision

Let expected gross commercial return be G_t , operating cost be c_t , insurance premium be Π_t , and expected loss be $E_t[L]$.

A stylized continuation condition is

$$E_t[G_{t+1}] > c_t + \Pi_t + \text{capital cost}_t. \quad (40)$$

The insurance premium transforms an uncertain physical loss into a more predictable financial expenditure. Risk is therefore not eliminated; it is priced and redistributed.

This parallels modern portfolio theory, in which investors trade expected return against risk [20], and asset-pricing theory, in which systematic risk affects required returns [28, 19].

9 Econometric Interpretation

The ten-ship identity can be interpreted empirically as a set of latent commercial transition probabilities.

Let

$$Y_{it} \in R, F, L \quad (41)$$

represent the outcome of voyage i at time t .

A multinomial model can specify

$$P(Y_{it} = j \mid X_{it}) = \frac{\exp(X'_{it}\beta_j)}{\sum_{k \in R, F, L} \exp(X'_{it}\beta_k)}. \quad (42)$$

The expected number of outcomes in a fleet of N ships is then

$$E[N_j \mid X] = NP(Y = j \mid X). \quad (43)$$

The empirical question is consequently not whether the exact ratio 8 : 1 : 1 is universally true. It is whether commercial systems exhibit persistent, measurable probabilities of realization, foreign persistence, and destruction.

Panel data, survival analysis, competing-risks models, duration models, and Markov transition models provide natural empirical frameworks [8, 32].

10 Statistics and Information

Let the state estimator be

$$\hat{X}_t = E[X_t \mid \mathcal{F}_t], \quad (44)$$

where $\mathcal{F}_t$ represents information available at time t .

Commercial decisions therefore depend not on the true state alone but on the agent's information about the state.

This introduces Bayesian learning:

$$P(\theta \mid D) \propto P(D \mid \theta)P(\theta), \quad (45)$$

where θ contains unknown parameters governing commercial return, persistence, and loss [5].

11 Psychology and Behavioral Economics

Commercial decisions are not necessarily generated by expected-value maximization.

Risk perception may depend on loss aversion, ambiguity, overconfidence, reference dependence, and framing [14, 15].

Let perceived risk be $\tilde{p}_t$ rather than objective risk p_t . Then

$$\tilde{p}_t \neq p_t \quad (46)$$

may generate systematic deviations from the theoretically optimal stopping rule.

The distinction between risk and ambiguity is particularly relevant:

$$\text{Risk} = \text{known or estimable probabilities}, \quad (47)$$

whereas

$$\text{Ambiguity} = \text{uncertainty concerning the probabilities themselves}. \quad (48)$$

Commerce therefore requires both insurance against measurable hazards and institutional mechanisms for coping with uncertainty.

12 Biology and Evolution

The ten-ship structure has an evolutionary analogue.

A population sends propagules into an uncertain environment. Some return benefits to the source population, some establish persistent populations elsewhere, and some fail.

Let w_R, w_F, w_L denote fitness contributions:

$$\bar{w} = rw_R + fw_F + \ell w_L. \quad (49)$$

Selection favors strategies that increase expected reproductive or productive value, subject to environmental constraints [10, 21].

The analogy is structural rather than literal: commerce, biological reproduction, and evolutionary adaptation all involve allocation of resources under uncertainty.

13 Physics and Thermodynamics

A physical interpretation emphasizes transformation rather than creation ex nihilo.

Let total accounting value be decomposed into productive, dissipative, and transferred components:

$$V_0 = V_{\text{productive}} + V_{\text{persistent}} + V_{\text{dissipated}}, \quad (50)$$

subject to the relevant accounting convention.

Thermodynamics distinguishes useful work from dissipation and establishes the irreversibility of many physical processes [25]. The analogy suggests a useful economic distinction between reversible exchange and irreversible destruction.

The sunk ship represents an irreversible physical event. The foreign ship represents relocation of productive capacity.

14 Chemistry

Chemical reaction networks provide another analogy. Reactants can be transformed into desired products, persistent intermediates, and waste.

Schematically,



where B represents realized commercial output, C persistent productive capacity, and D destruction or waste.

Reaction pathways depend on rates, activation barriers, catalysts, and environmental conditions. Commerce similarly depends on transaction costs, institutions, information, infrastructure, and trust.

15 Network Theory

Commerce is fundamentally a network phenomenon.

Let $G = (V, E)$ be a directed weighted graph whose nodes are economic jurisdictions and whose edges represent commercial relationships.

A persistent foreign ship corresponds to the creation or strengthening of an edge:

$$w_{ij,t+1} > w_{ij,t}. \quad (52)$$

Repeated commerce therefore creates network centrality, redundancy, dependence, and systemic exposure [11, 3].

The commercial system can consequently be represented by

$$\mathbf{W}_{t+1} = F(\mathbf{W}_t, \mathbf{X}_t, \boldsymbol{\varepsilon}_t). \quad (53)$$

16 International Relations and Diplomacy

Persistent commerce creates repeated interactions between political jurisdictions.

Trade can create mutual dependence, bargaining power, information channels, and institutional relationships. The political consequences of commercial interdependence have been studied extensively in international political economy [16, 17].

Let M_{ij} denote mutual dependence between nations i and j . A stylized relationship is

$$M_{ij} = f(T_{ij}, K_{ij}, F_{ij}), \quad (54)$$

where T_{ij} is trade, K_{ij} cross-border capital, and F_{ij} financial exposure.

Commerce therefore has diplomatic consequences without being reducible to diplomacy.

17 Geopolitics

Commercial routes depend on geography, chokepoints, maritime security, political stability, and access to infrastructure.

Let route risk be

$$\rho_{ij,t} = f(\text{distance, chokepoints, conflict, piracy, weather, institutions}). \quad (55)$$

Expected commercial value is decreasing in sufficiently severe route risk:

$$\frac{\partial E[V]}{\partial \rho} < 0. \quad (56)$$

Consequently, commerce generates incentives to secure routes, diversify suppliers, construct ports, and maintain diplomatic relations.

18 Warfare Economics

War changes the commercial state transition process.

Let W_t denote the intensity of armed conflict. Then the probability of maritime loss may be modeled as

$$p_t = p_0 + \alpha W_t, \quad \alpha > 0, \quad (57)$$

within a suitable range.

Expected commercial value becomes

$$E[V_{t+1}] = E[V_{t+1} | W_t]. \quad (58)$$

War therefore affects commerce through destruction, insurance costs, route closure, sanctions, capital flight, mobilization, and changes in expected returns.

This is one reason warfare economics cannot be separated completely from international commerce.

19 Welfare Economics

A commercial transaction need not maximize social welfare merely because it increases private profit.

Let private utility be U_P and social welfare be W_S . Then

$$W_S = U_P + E, \quad (59)$$

where E represents external effects.

Pollution, congestion, strategic dependence, labor displacement, knowledge spillovers, and systemic financial risk can produce

$$E \neq 0. \quad (60)$$

The social stopping rule may consequently differ from the private stopping rule.

Private continuation:

$$C_P > S_P. \quad (61)$$

Social continuation:

$$C_S > S_S. \quad (62)$$

The two need not coincide [23, 7].

20 Political Economy

Commerce requires property rights, contract enforcement, money, accounting, courts, political authority, and credible commitments.

Institutions reduce transaction costs:

$$TC = f(\text{information, enforcement, coordination, uncertainty}). \quad (63)$$

Lower transaction costs expand the set of commercially viable exchanges [6, 22, 31].

Thus the merchant's ten ships are embedded within a much larger institutional architecture.

21 Philosophy

The theory also contains a philosophical distinction between possession and relationship.

The returning ships represent realized possession. The foreign ship represents a relationship that continues beyond the current transaction. The sunk ship represents irreversibility.

Commerce therefore transforms three philosophical categories:

$$\text{Thing} \longrightarrow \text{realized thing, relationship, loss}. \quad (64)$$

Aristotelian distinctions between exchange, household management, and acquisition, later transformed by political economy, remain relevant to the conceptual distinction between useful exchange and accumulation [1, 30].

22 Religion and Moral Economy

Religious traditions have historically treated commerce through concepts including trust, fairness, debt, charity, stewardship, usury, and obligations to strangers.

The theoretical point is not to reduce commerce to theology but to recognize that commercial systems require normative expectations concerning acceptable conduct.

Repeated exchange is difficult to sustain when every participant expects deception.

Trust can therefore be treated as a state variable:

$$T_{t+1} = F(T_t, \text{honored contracts, breaches, sanctions}). \quad (65)$$

Commerce is consequently both an economic and a moral institution.

23 Computer Science and Algorithms

The optimal commercial decision can be implemented algorithmically.

Algorithm CommercialOptimalStopping

Input:

initial commercial capital K
state estimator X
discount factor β
stopping value S
transition model P
reward function R

Repeat:

observe information F_t
estimate state X_t
compute continuation value $C(X_t)$

'''

if $C(X_t) \leq S(X_t)$:

```

    STOP
else:
    insure exposed commercial capital
    execute commercial voyage
    observe return, persistence, or loss
    update X_t
    learn model parameters
'''

```

Output:
 stopping time tau*
 realized commercial wealth
 persistent foreign capital
 accumulated losses

The algorithm is a finite-state approximation to a Markov decision process [4, 26].

24 Machine Learning

A modern system can estimate the continuation value using reinforcement learning.

Let

$$Q(s, a) = E \left[\sum_{k=0}^{\infty} \beta^k R_{t+k} \mid s_t = s, a_t = a \right]. \quad (66)$$

The action set is

$$A = \text{continue, stop}. \quad (67)$$

The optimal policy is

$$\pi^*(s) = \arg \max_{a \in A} Q(s, a). \quad (68)$$

A neural approximation can represent

$$Q_{\theta}(s, a) \approx Q(s, a). \quad (69)$$

Relevant state variables could include freight rates, commodity prices, exchange rates, interest rates, insurance premia, geopolitical risk, weather, port congestion, vessel condition, and macroeconomic indicators.

Machine learning therefore provides a computational implementation of the theoretical stopping problem rather than replacing its economic structure.

25 Data Science Architecture

A practical empirical system can be organized into five layers:

1. data acquisition;
2. feature construction;
3. state estimation;
4. transition and risk modeling;
5. optimal stopping and policy evaluation.

$$\text{Raw Data} \rightarrow \text{Features} \rightarrow \text{State} \rightarrow \text{Forecast} \rightarrow \text{Decision}. \quad (70)$$

The system should separately estimate return probability, persistence probability, and loss probability:

$$\hat{p}_R = P(R \mid X), \quad (71)$$

$$\hat{p}_F = P(F \mid X), \quad (72)$$

$$\hat{p}_L = P(L \mid X). \quad (73)$$

The probabilities must satisfy

$$\hat{p}_R + \hat{p}_F + \hat{p}_L = 1. \quad (74)$$

26 A Unified Commercial Architecture

Figure 1 summarizes the theory.

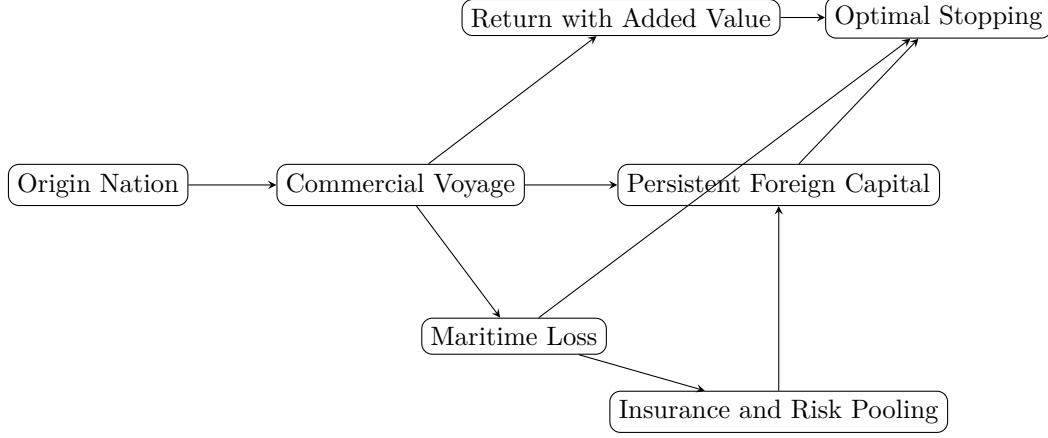


Figure 1: The commercial system as exchange, persistence, loss, insurance, and sequential decision-making.

27 The Core Propositions

Proposition 1 (Commercial Decomposition). *A commercial fleet can be decomposed into realized return, persistent foreign capital, and loss:*

$$N = N_R + N_F + N_L. \quad (75)$$

Proposition 2 (Persistence). *A commercial asset remaining abroad need not constitute economic loss. If it generates future productive services, its capitalized value is positive:*

$$V_F^* > V_F. \quad (76)$$

Proposition 3 (Non-Degenerate Insurance). *A commercial insurance market requires positive probability of loss:*

$$P(L > 0) > 0. \quad (77)$$

Theorem 1 (Commercial Optimal Stopping). *Let the commercial state process be Markovian and let the value function be finite. Then the optimal commercial strategy is characterized by*

$$W(x) = \max S(x), C(x), \quad (78)$$

and the optimal stopping time is

$$\tau^* = \inf t : C(X_t) \leq S(X_t). \quad (79)$$

Proof. The result follows from the Bellman principle of optimality. At each state the decision-maker compares immediate liquidation value with the expected discounted value of continuation. Choosing the larger value yields the Bellman equation, and the first entry into the stopping region defines the optimal stopping time [4, 26]. $\square$

28 A Compact Accounting Table

Table 1: The canonical ten-ship commercial system

Outcome	Ships	Fraction	Economic interpretation
Return	8	0.80	Realized exchange and added value
Persistence	1	0.10	Foreign commercial capital
Loss	1	0.10	Destruction and risk realization
Total	10	1.00	Initial commercial fleet

29 The General Commercial Equation

The central accounting identity is

$$N = N_R + N_F + N_L. \quad (80)$$

The corresponding economic identity is

$$\text{Commerce} = \text{Realized Exchange} + \text{Persistent Capital} - \text{Destruction}. \quad (81)$$

When insurance is introduced,

$$\text{Commerce} = \text{Exchange} + \text{Persistence} + \text{Risk} + \text{Risk Pooling}. \quad (82)$$

When intertemporal choice is introduced,

$$\text{Commerce} = \frac{\text{Stochastic}}{\text{Exchange}} + \frac{\text{Capital}}{\text{Accumulation}} + \frac{\text{Risk}}{\text{Management}} + \frac{\text{Optimal}}{\text{Stopping}}. \quad (83)$$

30 Limits of the Theory

The ten-ship ratio is a conceptual benchmark, not an empirical constant. Actual commercial systems differ across commodities, periods, technologies, political institutions, insurance arrangements, transport modes, and jurisdictions.

Furthermore, “value” must be defined carefully. Physical value, accounting value, market value, utility, social welfare, and strategic value are not identical.

The theory therefore should be tested rather than treated as an empirical law.

A suitable empirical program would estimate:

$$P(R | X), \quad (84)$$

$$P(F | X), \quad (85)$$

$$P(L | X), \quad (86)$$

$$(87)$$

estimate their determinants, calculate the value of persistence, estimate insurance premia and losses, and test whether observed stopping decisions are consistent with the predicted continuation boundary.

31 Conclusion

The real structure of commerce is richer than a simple exchange equation.

A commercial expedition begins with resources and enters an uncertain environment. Some value returns enhanced; some productive capital remains embedded in the foreign economy; some capital is destroyed. The possibility of destruction creates a demand for insurance. Persistent capital creates future commercial relationships. Repeated interaction creates networks, institutions, diplomatic relationships, and strategic dependencies. Finally, every commercial system confronts an intertemporal question: continue, or stop?

The canonical ten-ship representation captures the architecture:

$$\boxed{10 = 8 + 1 + 1.} \quad (88)$$

The first component is realization.

The second is persistence.

The third is loss.

Insurance converts the financial consequences of the third into a poolable risk. Optimal stopping determines whether another voyage is worth undertaking. International relations determine the political environment in which the voyage occurs. Welfare economics determines whether private continuation coincides with social continuation. Statistics and econometrics allow the transition probabilities to be estimated. Computer science provides algorithms for the decision process, while artificial intelligence can approximate the continuation value in sufficiently complex environments.

The resulting theory is therefore not a theory of ships alone. It is a theory of **resource movement through uncertain, persistent, institutional, and strategic networks**.

In its most compact form:

$$\boxed{\textit{Commerce is the stochastic transformation of resources through exchange, persistence, risk, and choice over time.}} \quad (89)$$

References

- [1] Aristotle. *Politics*. Translated by Benjamin Jowett. Batoche Books, 1999.
- [2] K. J. Arrow. *Essays in the Theory of Risk-Bearing*. Markham, 1971.
- [3] A.-L. Barabási and R. Albert. Emergence of scaling in random networks. *Science*, 286:509–512, 1999.
- [4] R. Bellman. *Dynamic Programming*. Princeton University Press, 1957.
- [5] J. O. Berger. *Statistical Decision Theory and Bayesian Analysis*. Springer, 1985.
- [6] R. H. Coase. The nature of the firm. *Economica*, 4:386–405, 1937.
- [7] R. H. Coase. The problem of social cost. *Journal of Law and Economics*, 3:1–44, 1960.
- [8] D. R. Cox. Regression models and life-tables. *Journal of the Royal Statistical Society: Series B*, 34:187–220, 1972.
- [9] J. H. Dunning. *Explaining International Production*. Unwin Hyman, 1988.
- [10] I. Fisher. *The Theory of Interest*. Macmillan, 1930.
- [11] M. Granovetter. Economic action and social structure: The problem of embeddedness. *American Journal of Sociology*, 91:481–510, 1985.
- [12] F. A. Hayek. *The Pure Theory of Capital*. University of Chicago Press, 1941.
- [13] S. H. Hymer. *The International Operations of National Firms: A Study of Direct Foreign Investment*. MIT Press, Cambridge, MA, 1976.
- [14] D. Kahneman and A. Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47:263–291, 1979.
- [15] D. Kahneman. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, 2011.
- [16] R. O. Keohane. *After Hegemony*. Princeton University Press, 1984.
- [17] R. O. Keohane and J. S. Nye. *Power and Interdependence*. Scott, Foresman, 1989.
- [18] P. Krugman. Scale economies, product differentiation, and the pattern of trade. *American Economic Review*, 70:950–959, 1980.
- [19] J. Lintner. The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. *Review of Economics and Statistics*, 47:13–37, 1965.

- [20] H. Markowitz. Portfolio selection. *Journal of Finance*, 7:77–91, 1952.
- [21] J. Maynard Smith. *Evolution and the Theory of Games*. Cambridge University Press, 1982.
- [22] D. C. North. *Institutions, Institutional Change and Economic Performance*. Cambridge University Press, 1990.
- [23] A. C. Pigou. *The Economics of Welfare*. Macmillan, 1920.
- [24] K. Polanyi. *The Great Transformation*. Farrar & Rinehart, 1944.
- [25] I. Prigogine. *Self-Organization in Nonequilibrium Systems*. Wiley, 1977.
- [26] M. L. Puterman. *Markov Decision Processes*. Wiley, 1994.
- [27] D. Ricardo. *On the Principles of Political Economy and Taxation*. John Murray, 1817.
- [28] W. F. Sharpe. Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance*, 19:425–442, 1964.
- [29] R. J. Shiller. *The New Financial Order*. Princeton University Press, 2003.
- [30] A. Smith. *An Inquiry into the Nature and Causes of the Wealth of Nations*. W. Strahan and T. Cadell, 1776.
- [31] O. E. Williamson. *The Economic Institutions of Capitalism*. Free Press, 1985.
- [32] J. M. Wooldridge. *Econometric Analysis of Cross Section and Panel Data*. MIT Press, 2010.

Glossary

Commerce The organized exchange and movement of resources, goods, services, capital, and claims among economic agents.

Commercial capital Resources deployed with the expectation of generating future commercial value.

Persistent foreign capital Productive capital that remains in a foreign jurisdiction after a commercial transaction or expedition.

Commercial loss Destruction or irreversible disappearance of commercial capital.

Risk Uncertainty for which relevant probabilities can be represented or estimated.

Ambiguity Uncertainty concerning the appropriate probability distribution itself.

Insurance An institutional mechanism for pooling and transferring financial consequences of uncertain losses.

Continuation value The expected discounted value of continuing an activity rather than terminating it.

Stopping value The value obtained by terminating the activity in the current state.

Stopping time A random time determined by information available up to that time.

Stopping region The set of states in which termination is optimal.

Continuation region The set of states in which continuation is optimal.

Commercial network A network of economic relationships linking firms, individuals, markets, and jurisdictions.

Transaction cost The cost of searching, negotiating, monitoring, enforcing, and maintaining an exchange.

Systemic risk Risk arising from interactions among components of a sufficiently interconnected economic system.

Commercial state The collection of variables sufficient to characterize the relevant condition of the commercial system at a given time.

Value function The maximal expected discounted value attainable from a given state under an optimal policy.

Bellman equation A recursive representation of an optimization problem based on the principle of optimality.

Policy A mapping from observed states to actions.

Expected loss The probability-weighted magnitude of an uncertain loss.

Welfare A measure of social economic well-being that may differ from private profit.

The End